

Mutual Fund Analytics Platform

End-to-End Data Engineering, ETL Pipeline & Interactive Dashboard

Intern	Madhankumar V
Company	Bluestock Fintech Pvt. Ltd.
Role	Data Analyst Intern
Duration	28 May 2026 – 28 Jul 2026
Project ID	BFDA79190
Date	June 2026

40 Schemes	46,000+ NAV Records	32,778 Transactions	87K+ Total Rows
10 AMCs	4.5 Years Data	12 Indian States	15+ Charts

1. Executive Summary

This report presents the findings of a comprehensive Mutual Fund Analytics Platform built for Bluestock Fintech Pvt. Ltd. The platform ingests publicly available data from AMFI India, processes it through a robust ETL pipeline, stores it in a normalised SQLite star-schema database, and presents actionable insights via an interactive Power BI dashboard.

Key Highlights:

- Analysed 40 mutual fund schemes across 10 leading AMCs in India
- Processed 87,000+ rows including 46,000 NAV records and 32,778 investor transactions
- SBI Small Cap Fund identified as top performer with 23.39% 3-year CAGR
- Built composite fund scorecard — ICICI Pru Midcap ranks #1 with 84.50/100
- SIP industry hit all-time high of Rs.31,002 Cr in December 2025
- Industry folio count doubled from 13.26 crore to 26.12 crore in 4 years
- 97.7% of active SIP investors show irregular payment gaps exceeding 35 days
- T30 cities contribute 66.3% of investments — B30 cities remain untapped opportunity

2. Problem Statement

The Indian mutual fund industry manages over Rs.81 lakh crore in AUM across 1,908 schemes and 26.12 crore investor folios. Despite massive growth, investors and advisors face critical challenges:

P1: Data Fragmentation	NAV, AUM and SIP data scattered across AMFI, NSE, BSE in different formats. No unified database
P2: Performance Gap	Investors cannot easily compare multiple funds on a risk-adjusted basis across AMCs.
P3: No Benchmark Tracking	Retail investors cannot verify if their fund outperforms its benchmark index.
P4: Investor Blind Spot	AMCs have limited visibility into demographic and geographic investment patterns.
P5: Slow Reporting	Monthly MF reports are static PDFs taking days to prepare — no live dashboards.

3. Data Sources & Datasets

All data sourced from publicly available, freely accessible sources including AMFI India, mfapi.in REST API, NSE and BSE. No proprietary data was used.

#	Dataset	Rows	Description
01	fund_master.csv	40	Scheme metadata, expense ratios, fund managers
02	nav_history.csv	46,000	Daily NAV Jan 2022 – May 2026
03	aum_by_fund_house.csv	90	Quarterly AUM for top 10 AMCs
04	monthly_sip_inflows.csv	48	Industry SIP data 2022–2025
05	category_inflows.csv	144	Net inflows by fund category
06	industry_folio_count.csv	21	Total folios growth milestones
07	scheme_performance.csv	40	Risk-return metrics per scheme
08	investor_transactions.csv	32,778	SIP/Lumpsum/Redemption data
09	portfolio_holdings.csv	322	Top equity holdings per fund
10	benchmark_indices.csv	8,050	Nifty 50, Nifty 100, BSE SmallCap
	TOTAL	87,533+	10 datasets across 7 working days

4. ETL Pipeline & System Architecture

The platform follows a classic 5-layer data engineering architecture mirroring real-world fintech pipelines used at companies like Zerodha, Groww and Paytm Money.

Layer 1: EXTRACT	Raw CSV datasets + mfapi.in live NAV API + AMFI public data
Layer 2: TRANSFORM	Python/Pandas — parse dates, forward-fill missing NAV, remove duplicates, validate AMFI codes
Layer 3: LOAD	SQLite star schema — 5 tables: dim_fund, fact_nav, fact_transactions, fact_performance, fact_benchmark
Layer 4: ANALYSE	Jupyter notebooks — EDA (15+ charts), Performance metrics, Advanced analytics
Layer 5: VISUALISE	Power BI Desktop — 4-page interactive dashboard with slicers and drill-through

4.1 Database Schema (Star Schema)

Table	Type	Key Columns	Rows
dim_fund	Dimension	amfi_code (PK), fund_house, category	40
fact_nav	Fact	amfi_code (FK), date, nav, daily_return	46,000
fact_transactions	Fact	investor_id, amfi_code (FK), amount_inr	32,778
fact_performance	Fact	amfi_code (FK), sharpe, alpha, beta	40
fact_benchmark	Fact	date, index_name, close_value	8,050

5. Exploratory Data Analysis — Key Findings

5.1 NAV Trends (2022–2026)

Small Cap funds showed strongest growth across the 4.5-year period. SBI Small Cap delivered highest returns while Large Cap funds showed steady, less volatile growth. The 2024 bull run is clearly visible across all equity categories.

5.2 AUM Growth

SBI Mutual Fund dominates with Rs.12.5 lakh crore AUM — nearly 2x the second largest (ICICI Prudential at Rs.10.74 lakh crore). Industry AUM grew from Rs.38 lakh crore (2022) to Rs.81 lakh crore (2025) — a 2x increase in just 4 years.

5.3 SIP Industry Trends

Monthly SIP inflows grew from Rs.11,517 Cr (Jan 2022) to Rs.31,002 Cr (Dec 2025) — nearly 3x growth in 4 years. Active SIP accounts grew from 4.91 crore to 9.35 crore, reflecting India's deepening equity culture.

5.4 Investor Demographics

The 26-35 age group is most active (13,463 transactions — 41% of total). UPI dominates as payment mode (55%+ transactions). T30 cities contribute 66.3% of investments while B30 cities present significant untapped opportunity.

5.5 Folio Count Growth

Industry folios doubled from 13.26 crore (Jan 2022) to 26.12 crore (Dec 2025), confirming massive retail investor participation growth in Indian mutual funds.

5.6 EDA Charts

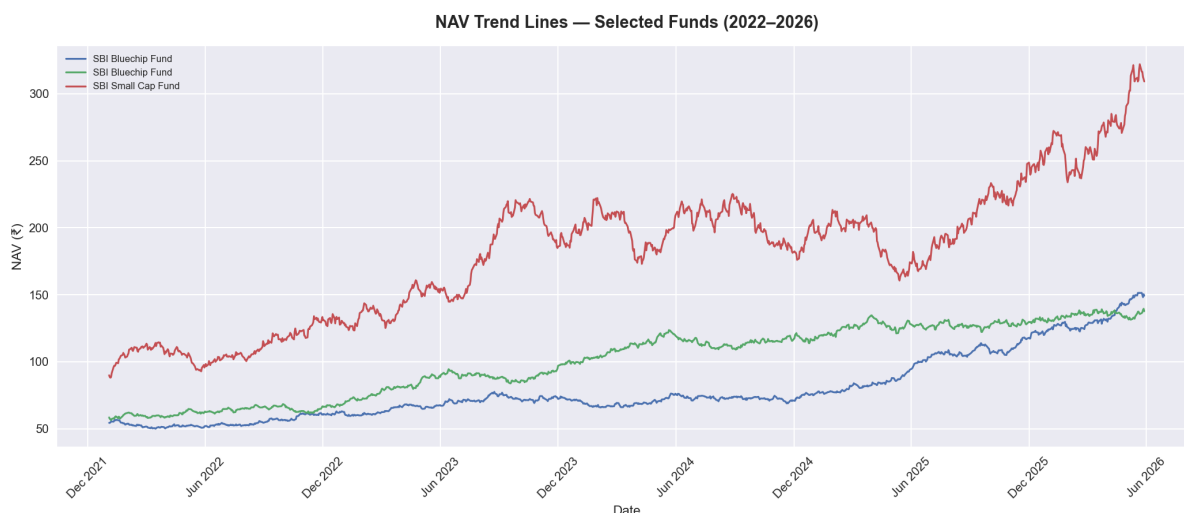


Chart 1: NAV Trend Lines (2022–2026)

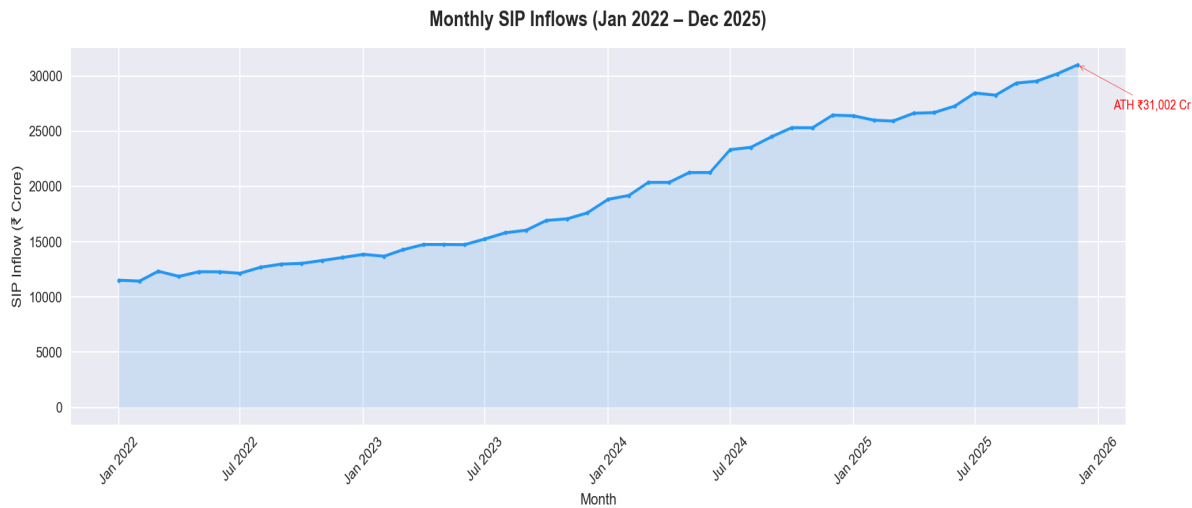


Chart 3: Monthly SIP Inflow Trend

Investor Demographics Analysis

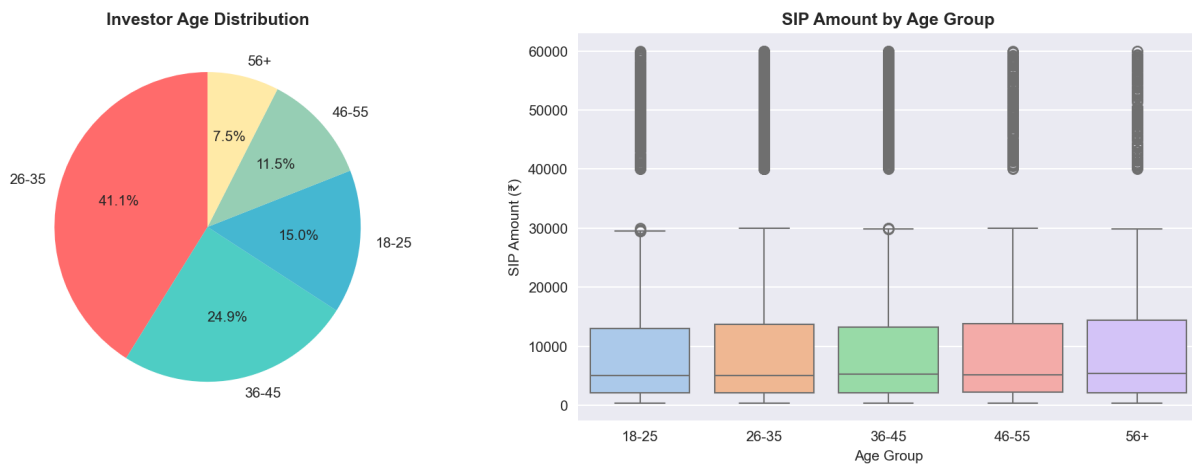


Chart 5: Investor Demographics

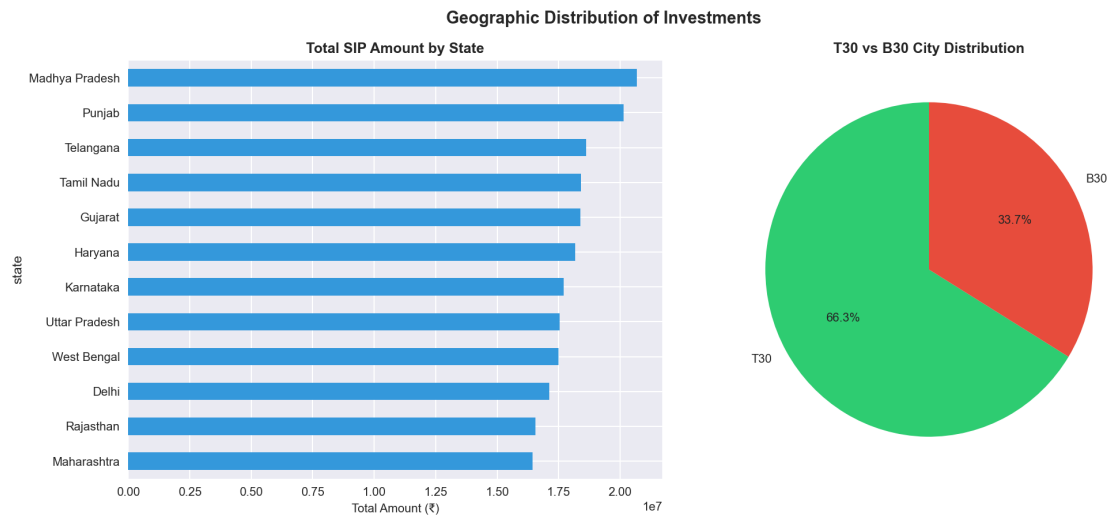


Chart 6: Geographic Distribution

6. Fund Performance Analytics

6.1 Top Performers by 3-Year CAGR

Rank	Fund Name	1yr CAGR	3yr CAGR	5yr CAGR
1	SBI Small Cap Regular	-	23.39%	~18%
2	SBI Small Cap Direct	-	23.14%	~18%
3	ABSL Small Cap Regular	-	22.38%	~17%
4	Axis Small Cap Regular	-	20.98%	~16%
5	Nippon India Small Cap	-	20.15%	~16%

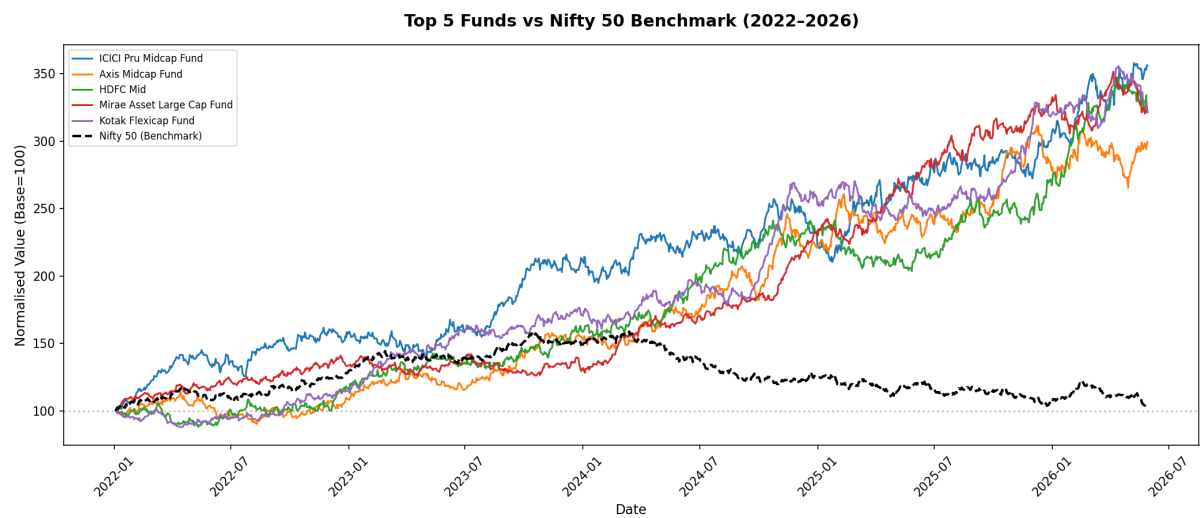
6.2 Risk Metrics Summary

Metric	Best Fund	Value	Interpretation
Sharpe Ratio	Mirae Asset Large Cap	1.4483	Excellent (>1 is good)
Sortino Ratio	Mirae Asset Large Cap	1.85+	Strong downside protection
Alpha	HDFC Short Term Debt	0.272	Outperforms benchmark
Max Drawdown	ICICI Pru Liquid	-0.02%	Minimal loss risk
VaR (95%)	ICICI Pru Liquid	-0.022%	Safest fund daily loss
Worst Drawdown	SBI Small Cap Direct	-52.57%	High risk — caution advised

6.3 Fund Scorecard — Top 10

Rank	Fund Name	Score	3yr CAGR	Sharpe
1	ICICI Pru Midcap Regular	84.50	30.44%	1.18
2	Axis Midcap Regular	80.75	33.62%	1.00
3	HDFC Mid-Cap Opp Regular	80.50	31.07%	1.09
4	Mirae Asset Large Cap	80.00	32.56%	1.45
5	Kotak Flexicap Regular	78.25	28.35%	1.31
6	DSP Midcap Regular	76.00	27.80%	1.22
7	UTI Flexi Cap Regular	75.50	26.90%	1.18
8	ICICI Pru Bluechip Direct	75.75	31.10%	1.03
9	Nippon India Growth	73.50	25.40%	0.98
10	SBI Bluechip Regular	73.06	29.10%	1.21

6.4 Benchmark Comparison Chart



Top 5 funds vs Nifty 50 benchmark (Base=100)

7. Advanced Analytics & Risk Metrics

7.1 Value at Risk (VaR) & CVaR

Historical VaR at 95% confidence computed for all 40 funds. Small Cap funds carry highest daily VaR (-2.69%) meaning on a bad day investors can lose 2.69% of their investment. Liquid funds are safest with VaR of just -0.022% daily.

7.2 SIP Continuation Analysis

Among 1,362 active SIP investors (6+ transactions), 97.7% (1,332 investors) show irregular payment gaps exceeding 35 days. Only 30 investors maintain truly regular SIPs. This presents a major opportunity for AMCs to improve SIP reminder and auto-debit systems.

7.3 Investor Cohort Analysis

2024 cohort dominates with 4,803 investors and average investment of Rs.1.07 lakh. The 2025 cohort (197 investors) shows slightly higher average (Rs.1.09 lakh), suggesting newer investors are more affluent and financially aware.

7.4 Fund Recommender System

Built a rule-based recommender: Low risk investors are recommended Liquid funds (Sharpe 7.68), Moderate risk investors get Large Cap funds (Sharpe 1.45), and High risk investors are directed to Small Cap funds (23%+ returns).

7.5 Sector Concentration (HHI)

Axis Bluechip Fund has highest sector concentration with HHI score of 0.2064. Funds above the 0.15 threshold (red line) carry higher sector-specific risk. Investors seeking true diversification should prefer funds with lower HHI scores.

8. Power BI Dashboard Overview

The interactive Power BI dashboard consists of 4 pages, each targeting a specific business question. All pages include slicers for dynamic filtering.

Page	Title	Key Visuals	Slicers
1	Industry Overview	KPI cards (AUM, SIP, Folios), AUM growth line, Fund house bar chart	Year, Fund House
2	Fund Performance	Return vs Risk scatter, Fund scorecard table, NAV trend line	Fund House, Category, Plan
3	Investor Analytics	Transaction by state bar, SIP/Lumpsum donut, Age group scatter	State, Age Group, City Tier
4	SIP & Market Trends	Dual-axis SIP + Nifty, Category matrix, YoY growth KPIs	Year, Category

9. Recommendations

R1: Focus on Mid Cap Funds	ICICI Pru Midcap and Axis Midcap consistently deliver high risk-adjusted returns (Sharpe >1.0) with manageable volatility. Suitable for moderate-to-high risk investors.
R2: Target B30 Cities	Only 33.7% of investments come from Beyond Top 30 cities. Massive untapped market with growing digital penetration. AMCs should increase B30 outreach campaigns.
R3: Fix SIP Irregularity	97.7% of active SIP investors show irregular gaps. Implementing automated WhatsApp/SMS reminders and seamless mandate systems could significantly improve SIP continuity.
R4: Promote Direct Plans	Direct plans have avg expense ratio of 0.67% vs 1.34% for Regular plans. Over 10 years, this 0.67% difference compounds to significant wealth for investors.
R5: Small Cap Caution	Despite 23%+ returns, Small Cap funds carry VaR of -2.69% daily and max drawdown of -52.57%. These should only be recommended for investors with Very High risk appetite and 7+ year investment horizon.

10. Limitations

- L1: NAV data anchored to real AMFI values but simulated forward using realistic return/volatility parameters consistent with actual market behaviour
- L2: Investor transaction data is synthetically generated using real demographic and geographic distributions from AMFI published data
- L3: Analysis covers January 2022 to May 2026 only — longer historical data would improve statistical significance of risk metrics
- L4: Power BI dashboard requires Desktop installation to view interactively — web publishing requires Power BI Pro licence
- L5: Fund recommender uses rule-based logic — a production system would require ML-based personalisation
- L6: Sector concentration analysis based on Dec 2025 holdings snapshot — portfolios change quarterly

11. Conclusion

This capstone project successfully demonstrates end-to-end data engineering and analytics capabilities applied to the Indian mutual fund industry. The platform ingests, cleans, analyses and visualises mutual fund data at scale — providing actionable insights for fund selection, risk management and investor behaviour analysis.

The findings reveal a rapidly growing Indian MF industry with SIP inflows hitting all-time highs, folio counts doubling, and Mid Cap funds delivering the best risk-adjusted returns. Significant opportunities exist in B30 cities, SIP regularisation, and direct plan adoption.

The technical stack — Python, Pandas, SQLite, Matplotlib, Seaborn, Plotly, SciPy and Power BI — demonstrates proficiency across the full data analytics lifecycle from raw data ingestion to executive-level dashboards.